
Basic DOS Commands in Windows

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1. Introduction

Documentation for first major labwork on CLI in Windows

2. Commands and Outputs

Task 1: Directory creation and verification

```
cd PHY134SEM12026
mkdir CLI_DOS_July2026
cd CLI_DOS_July2026 dir /p
```

```
Microsoft Windows [Version 10.0.26200.8973]
(c) Microsoft Corporation. All rights reserved.

C:\Users\HP>mkdir PHY134SEM12026

C:\Users\HP> cd PHY134SEM12026

C:\Users\HP\PHY134SEM12026> mkdir CLI_DOS_July2026

C:\Users\HP\PHY134SEM12026>cd CLI_DOS_July2026

C:\Users\HP\PHY134SEM12026\CLI_DOS_July2026> dir /p
Volume in drive C has no label.
Volume Serial Number is E03D-621F

Directory of C:\Users\HP\PHY134SEM12026\CLI_DOS_July2026

03/08/2026  20:16    <DIR>          .
03/08/2026  20:16    <DIR>          ..
               0 File(s)                0 bytes
               2 Dir(s)  85,903,147,008 bytes free
```

Task 2: File creation

```
echo CLI DOS Experiment Notes > notes.txt
type notes.txt
more notes.txt
```

```
C:\Users\HP\PHY134SEM12026\CLI_DOS_July2026>echo CLI DOS Experiment Notes > notes.txt
C:\Users\HP\PHY134SEM12026\CLI_DOS_July2026>type notes.txt
CLI DOS Experiment Notes
C:\Users\HP\PHY134SEM12026\CLI_DOS_July2026>more notes.txt
CLI DOS Experiment Notes
```

Task 3: File Relocation

```
mkdir Data move notes.txt Data
cd Data dir
cd ..
```

```
C:\Users\HP\PHY134SEM12026\CLI_DOS_July2026>mkdir Data

C:\Users\HP\PHY134SEM12026\CLI_DOS_July2026>mv notes.txt Data
'mv' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\HP\PHY134SEM12026\CLI_DOS_July2026>move notes.txt Data
1 file(s) moved.

C:\Users\HP\PHY134SEM12026\CLI_DOS_July2026>cd Data

C:\Users\HP\PHY134SEM12026\CLI_DOS_July2026\Data>dir
Volume in drive C has no label.
Volume Serial Number is E03D-621F

Directory of C:\Users\HP\PHY134SEM12026\CLI_DOS_July2026\Data

03/08/2026  20:21    <DIR>          .
03/08/2026  20:21    <DIR>          ..
03/08/2026  20:19                27 notes.txt
                   1 File(s)                27 bytes
                   2 Dir(s)  85,803,954,176 bytes free

C:\Users\HP\PHY134SEM12026\CLI_DOS_July2026\Data>cd ..

C:\Users\HP\PHY134SEM12026\CLI_DOS_July2026>|
```

Task 4: System Diagnostics Script

```
type system_report.txt
```

```

C:\Users\HP\PHY134SEM12026\CLI_DOS_July2026>type system_report.txt
System Information Report
KojagorisLaptop

Windows IP Configuration

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 10:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    IPv6 Address. . . . . : 2405:201:8016:e86b:57a:8104:e6c3:3f18
    Temporary IPv6 Address. . . . . : 2405:201:8016:e86b:38e8:a9e5:1e15:6717
    Link-local IPv6 Address . . . . . : fe80::2f4a:f42b:90db:7dbb%9
    IPv4 Address. . . . . : 192.168.29.167
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::b6a7:c6ff:fe4d:e95d%9
                                192.168.29.1

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :
Name
Intel(R) Core(TM) i7-10870H CPU @ 2.20GHz
Report generated on 03/08/2026 at 20:24:41.45

C:\Users\HP\PHY134SEM12026\CLI_DOS_July2026>|

```

Task 5: Txt File Creation and update

```

echo Test File > test.txt
echo Updated on %DATE% >> test.txt
cls

```

```

C:\Users\HP\PHY134SEM12026\CLI_DOS_July2026>echo Test File > test.txt
C:\Users\HP\PHY134SEM12026\CLI_DOS_July2026>echo Updated on %DATE% >> test.txt
C:\Users\HP\PHY134SEM12026\CLI_DOS_July2026>notepad test.txt
C:\Users\HP\PHY134SEM12026\CLI_DOS_July2026>cls|

```

Task 6: sin(x) Plotting

```

cd gnuplot
"C:\Program Files\gnuplot\bin\gnuplot.exe"

```

```
plot sin(x)
```

```
C:\Users\HP>cd PHY134SEM12026\CLI_DOS_July2026\gnuplot
C:\Users\HP\PHY134SEM12026\CLI_DOS_July2026\gnuplot>wgnuplot
'wgnuplot' is not recognized as an internal or external command,
operable program or batch file.
C:\Users\HP\PHY134SEM12026\CLI_DOS_July2026\gnuplot>"C:\Program Files\gnuplot\bin\gnuplot.exe"

  G N U P L O T
  Version 6.0 patchlevel 4   last modified 2025-12-18

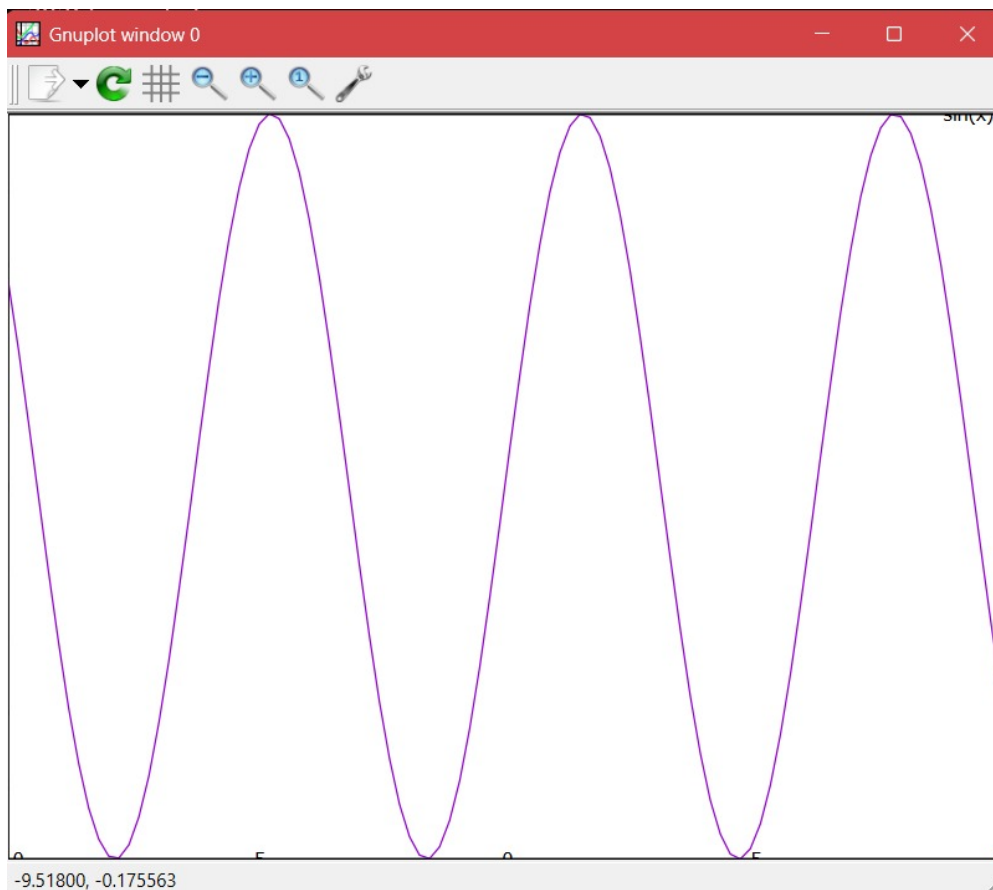
  Copyright (C) 1986-1993, 1998, 2004, 2007-2025
  Thomas Williams, Colin Kelley and many others

  gnuplot home:      http://www.gnuplot.info
  faq, bugs, etc:   type "help FAQ"
  immediate help:    type "help" (plot window: hit 'h')

  Terminal type is now qt
Encoding set to 'cp1252'.
gnuplot> plot sin(x)

Warning: slow font initializationgnuplot> qt_processTermEvent received a GE_fontprops event. This should not have happened
quit
C:\Users\HP\PHY134SEM12026\CLI_DOS_July2026\gnuplot>
```

Graph:



Task 7: Directory listing

```
dir /b > bare\_list.txt
more bare\_list.txt
```

```
C:\Users\HP\PHY134SEM12026\CLI_DOS_July2026>dir /b > bare_list.txt  
C:\Users\HP\PHY134SEM12026\CLI_DOS_July2026>more bare_list.txt  
bare_list.txt  
Data  
gnuplot  
system_report.txt  
test.txt  
C:\Users\HP\PHY134SEM12026\CLI_DOS_July2026>|
```

3. Conclusion

CLI and Batch Files

Using CLI for running commands is way faster than using GUI and essentially more streamlined when the number of files and folders is large and we need to run some commands like finding certain keywords or force delete something.

Gnuplot

Gnuplot is lightweight and fast for plotting purposes, surpassing standard methods like using Python (Matplotlib) for basic graphic and visualisation purposes.